

THE ACQUISITION OF SYNTACTIC CATEGORIES



CASE STUDY: THE ACQUISITION OF AUXILIARY AND LEXICAL VERBS

- Strange situation of syntax acquisition:
 - Unless children know what they have to learn before they begin, they cannot succeed in learning the grammar of their language.
- One might argue that these innate mechanisms are not specifically linguistic.
- Acquisition of auxiliary and lexical verbs used to determine whether children have specifically linguistic innate mechanisms or not.

LEXICAL VS AUXILIARY VERBS

- A lexical verb is the main verb of the sentence:
 - is the main action the subject is engaged in
 - Belongs to an indefinitely large group
- An auxiliary verb is supplemental verb that is “helping” the main verb
 - Comes from a fixed set
 - Primary auxiliaries: *do, have, and be*
 - Modal auxiliaries: *can (could), may (might), should (shall), will (would), must, need (also ought to, dare)*

SYNTACTIC SIMILARITIES BETWEEN THEM

- Follow subject noun phrases, lack grammatical properties of nouns, adjectives, etc.
- Some auxiliaries can occur as lexical verbs:
 - *She is swimming* vs *She is tall* (copula)
 - *She doesn't like sushi* vs *She did two exercises for homework*
 - *She had finished by six* vs *She has a Porsche.*

RESTRICTIONS ON AUXILIARIES

- Modal auxiliaries require that verb that follows is an infinitival form (*I can eat*)
- Perfect *have* requires perfect participle (*I have eaten*)
- Progressive *be* requires progressive participle (*I am eating*)
- Passive *be* requires passive participle (*I am eaten*)
- ⇒ How can children distinguish between auxiliary and lexical verbs before they learn the behavior of the two types?
- ⇒ How do they learn the two types of verbs' behavior before they can distinguish between them?

AUXILIARY VS LEXICAL VERB ERRORS

- Children overregularize lexical *be*, *do* and *have*, but never the auxiliary *be*, *do* and *have*
 - *She beed happy* but not *she beed smiling*
 - They treat the 2 types of verb differently
- Possible explanation:
 - Innate learning mechanism that leads to different handling of auxiliaries and lexical verbs
- But:
 - Not all languages have auxiliaries

ERRORS = CHILDREN ARE NOT PERFECT

- They make interesting errors:
 - Overgeneralize inflectional endings:
 - *Eated, mouses, ...*
 - Syntactic errors (e.g., with *do*):
 - *What she eats?*
- ⇒ children use language productively (not repeating what they heard)
- ⇒ *do*-support is not part of UG, but is a peculiar property of English

OTHER POSSIBILITY: FUNCTIONAL VS LEXICAL

- Lexical categories:
 - Nouns, verbs, adjectives, ...
 - They freely admit new members (fax, ...)
 - Grammatical behavior of one member of category can be generalized to other member of same lexical category
- Functional categories:
 - determiners, auxiliaries, pronouns, ...
 - New members are not welcome
 - Generalizations between members is dangerous

FUNCTIONAL/LEXICAL-DISTINCTION

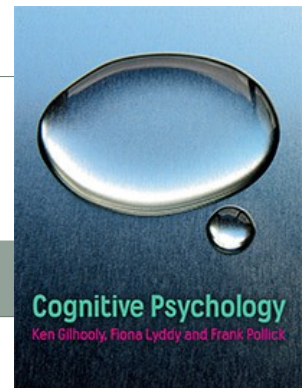
- Present in all human languages
 - Better candidate for being innate
- Other research (ERP, fMRI, ...) *sometimes* points to the importance of this distinction
- Helps to distinguish between:
 - Auxiliary/lexical verbs, pronouns/nouns, determiners/adjectives, ...
- Explains overgeneralization with lexical *be* and not with auxiliary *be*

IS TYPICAL OF DEBATE

- Do we need innateness?
- First answer = YES, and some proposal
- Next, not present in all languages
- Next, new, smaller proposal
- Next
- See Chomsky

TRANSFORMATIONAL GRAMMAR OF CHOMSKY

- Standard Theory, Extended Standard Theory, Revised Extended Standard Theory, Relational grammar, Minimalist Program
- First version:
 - Universal grammar program operated on nice chunks of language, such as noun phrases and verb phrases
 - But, e.g. some native Australian languages (Walpiri) no neatly packaged noun and verb phrases
 - See also Basque, Urdu as exceptions
- Last version:
 - Recursion is crucial aspect of language
 - Ability arose from single genetic mutation
 - However, some languages (Piraha) no recursion
- ...



WHAT ABOUT WORDS AND THEIR MEANING?

- Very common claim:
 - Languages grow up independently of each other
 - with no intrinsic meaning in the noises which form words
- Simple exception:
 - Onomatopoeias: a word that phonetically imitates or resembles the source of the sound that it describes
 - E.g., cuckoo, bam, ding-dong, drip, zipper, ...
- But recently:
 - Analysis of 40-100 basic vocabulary words in around 3,700 languages (almost 2/3 of the world's current languages).

WHAT ABOUT WORDS AND THEIR MEANING?

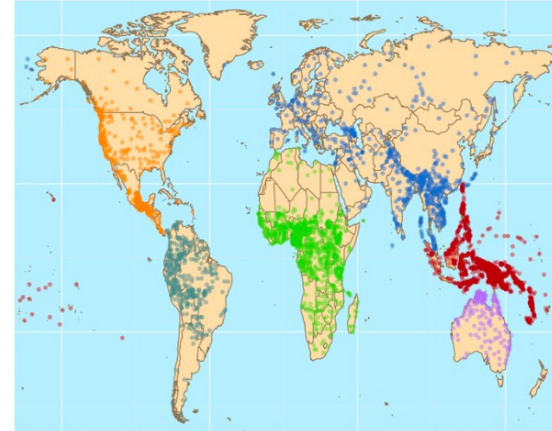


Fig. 1. Geographic distribution of the 6,452 word lists from the ASJP database (25). Colors distinguish different linguistic macroareas, regions with relatively little or no contact between them (but with much internal contact between their populations). These are North America (orange), South America (dark green), Eurasia (blue), Africa (green), Papua New Guinea and the Pacific Islands (red), and Australia (fuchsia).

WHAT ABOUT WORDS AND THEIR MEANING?

Concept	Positive symbol	Negative symbol
Ash	u	—
Bite	k	—
Bone	k	y
Breasts	u m	a h r
Dog	s	t
Drink	—	a
Ear	k	—
Eye	—	a
Fish	a	—
Full	p b	—
Hear	N	—
Horn	k r	—
I	s	u p b t s r l
Knee	o u p k q	—
Leaf	b p l	—
Name	i	o p
Nose	u n	a
One	t n	—
Red	r	—
Round	r	—
Sand	s	—
Skin	—	m n
Small	i c	—
Star	z	—
Stone	t	—
Tongue	e E l	u k
Tooth	—	b m
Water	—	t
We	n	p l s
You	—	u o p t d q s r l

Positive and negative signals are those that have frequency significantly larger and smaller than expected.

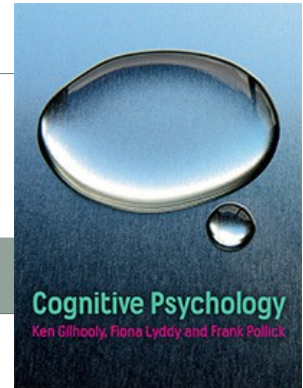
WHAT ABOUT WORDS AND THEIR MEANING?

- In most languages, the word for:
 - 'nose' is likely to include the sounds 'neh' or the 'oo' sound, as in 'ooze.'
 - 'leaf' is likely to include the sounds 'l,' 'p' or 'b'
 - 'sand' will probably use the sound 's'.
 - 'red' and 'round' are likely to include the 'r' sound
 - ...
- Why do humans tend to use the same sounds across languages to describe basic objects and ideas?

WHAT ABOUT WORDS AND THEIR MEANING?

- Danger of unfounded theories
- Authors (Blasi et al, 2016, PNAS) at a conference:
 - *Why “fire” with sound of “t”?*,
 - Everytime people came up with a reasonable explanation
 - But none of the given associations were true associations!

A CRITICAL PERIOD?



PIDGIN AND CREOLE LANGUAGES

- 17/18e century: plantation owners preferred to linguistically mix their slave populations
- ⇒ inter-language arises:
 - most often the language of plantation owners to be dominant
 - it's nobody's mother tongue
 - pidgin language

CREOLE LANGUAGES

- Difference between pidgin and creole language is acquisition:
 - Children of slaves frequently exposed to a pidgin language (= language of street)
 - Becomes primary language input
 - It is acquired as their mother tongue
- Transformation is more than terminological:
 - Pidgin is very much an incomplete language
 - Creole, as by miracle, has all that it takes to be a full-fledged natural human language (embedded clauses, relative clauses, consistent word order, ...)
- Second-generation pidgin-speakers “invent” a language that is more complex than the input (Jamaica, Haiti, Mauritius, Réunion, Hawaii, Pitcairn, etc.)

IDEA OF CRITICAL PERIOD/SENSITIVE PERIOD

- In many animal species, failure to learn various skills before a certain age makes it difficult or even impossible to learn those skills later:
 - Ducklings: ability to identify and follow the mother
 - Kittens: ability to perceive visual images.
 - Sparrows: ability to learn the father's song.
- "Sensitive period": biologically determined period during which learning must occur for development to *most likely* happen correctly

- Ideal experiment:
 - deprive children of all linguistic input during the purported critical period
 - and see how language development occurs.
- Is not very ethical 🙅
- However, deaf children, wild children, and second language learning

DEAF ISOLATES

- Deaf children born to hearing parents (without sign language):
 - No linguistic input
 - Interesting: individuals have normal early childhood experience, except for lack of language input
- As infants and toddlers, they seem to achieve the same early language milestones as hearing children:
 - first hand motions (cfr babbling), first real signs at same age as first words, short phrases with these signs cfr multi-word phrases

DEAF ISOLATES

- Newport and colleagues studied signing abilities for people whose first exposure to ASL (=American Sign Language) was at birth, before age 6 and after age 12
- Even after 30 years of using ASL:
 - Native signers outperform early signers
 - Early signers outperform late signers

WILD CHILDREN

- Skuse: reviewed 9 cases of children raised in conditions of extreme social and linguistic deprivation for 2,5-12 years
 - 6 eventually acquired (near-)normal language
 - Discovered by age 7
 - No signs of brain damage
 - The other 3:
 - One discovered at 5, but clear brain damage
 - One discovered at 3,5, but organic abnormalities
 - Genie, discovered at 14: language in right hemisphere

MORE ABOUT GENIE

- 1970s:
 - 13-year-old Genie brought to social services by her mother
 - Escaped together from mentally ill father
 - until mother's escape, she had no language input and horrific living conditions
- By age 17:
 - she had a 5-year-old's vocabulary
 - could express meanings by combining words together.

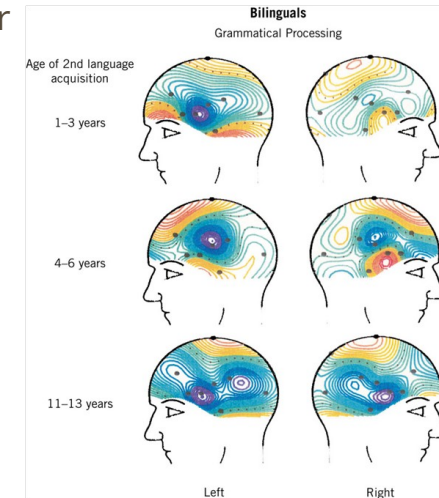
SLA (= SECOND LANGUAGE ACQUISITION)

- However, syntactic skills lagged far behind - deficient in both production and comprehension:
 - *"Mama wash hair in sink."; "Like go ride yellow school bus."; "At school scratch face."; "I want Curtiss play piano."*
 - So, "bad" language and only in the right hemisphere (in contrast with the "normal" observation of specialization in the left hemisphere):
 - Because no normal language acquisition until after puberty?
 - Because of cortical abnormalities?
- Oyama (1976): testing Italian immigrants learning English
 - age of arrival was better predictor of accent
 - than how many years the immigrant had been speaking English
 - Oyama (1978):
 - age of arrival was better predictor of comprehension
 - than number of years speaking the language
 - So not just about motor skill learning ability!

SLA (= SECOND LANGUAGE ACQUISITION)

- English abilities of native speakers of Korean and Chinese
 - Learning English before puberty:
 - Age of English immersion correlated highly with proficiency with English syntax and morphology
 - Learning English after puberty:
 - No such correlation
- In sum, the ability to acquire language diminishes with age

- Less hemispheric specialization occurs when a second language is learned at 4 years or older than when it is learned earlier



SOME NUANCES

- If there is a truly a critical period of language acquisition:
 - people learning language after this period should not succeed very well at all
 - people within the critical period should do very well.
- Hence, expectation is :
 - discontinuous function of performance



- However, evidence (Hakuta, Bialystok, & Wiley 2003) suggests that there is a smoother drop-off, and also a relation to education-level.
- = support for a *sensitive* instead of a *critical* period

OTHER EXPLANATIONS

- “Less is more” hypothesis (Newport, 1991):
 - Children can remember less than adults and have other cognitive limitations, like less attention.
 - Perhaps language is actually easier to figure out if the input is limited to smaller chunks.
 - Adults remember more and can store longer chunks, which makes their analytical task harder.
- Supported by connectionist computer simulation (Elman, 2001):
 - Better LA with shorter sentences as input

- Dominant language switch hypothesis:
 - Younger children are better able to make the new language their dominant language
 - Hence, better at new language than their old!
 - Jia & Aaronson (2003):
 - Grammatically judgment task English & Chinese
 - To native Chinese immigrants, to US between 1-38 y/o
 - Younger arrivals better on test English, but worse on test Chinese
 - Hence, better SLA because they switch to SL as their dominant language?

EARLY LEARNING LEAVES A BRAIN-TRACE (KLEIN ET AL, 2014)

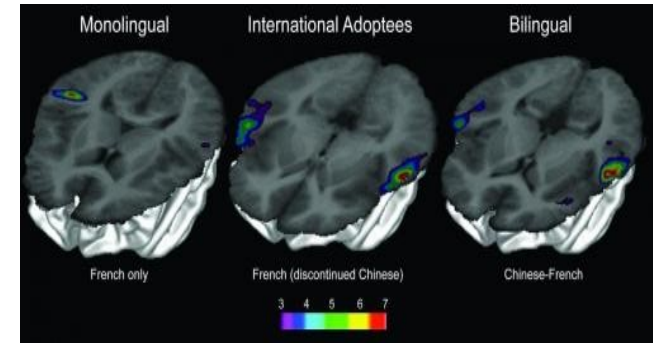
- Self-consciousness:
 - Less self-conscious about making errors, better performance
- Identification with new country/language:
 - Those who identified with new country/language, greater mastery
- Both “not being self-conscious” and identification more likely to be true in younger immigrants

- The language that an infant hears starting at birth
 - creates neural patterns that the brain retains years later
 - even if the child completely stops using the language.
- Experiment with three groups:
 - Chinese-French bilinguals
 - Adopted chinese (no chinese after adoption)
 - French monolinguals

EARLY LEARNING LEAVES A BRAIN-TRACE (KLEIN ET AL, 2014)

- 3 groups: a Chinese tonal task, differentiating a tone
- After training, everybody could do it equally well.
- But, differences in scans:
 - the brain activation pattern of the adopted Chinese who “lost” the language, matched the brain activation patterns for those who continued speaking Chinese since birth
 - and was completely different from the group of monolingual French speakers.

EARLY LEARNING LEAVES A BRAIN-TRACE (KLEIN ET AL, 2014)



- Hence, the infant brain developed Chinese language patterns at birth and never forgot them, even though the child no longer *speaks* or *understands* the language.